

Q.1. A dry cleaner uses exponential smoothing to forecast equipment usage at its main plant. August usage was forecasted to be 88 percent of capacity; actual usage was 89.6 percent of capacity. A smoothing constant of 0.1 is used.

A. Prepare a forecast for September.

B. Assuming actual September usage of 92 percent, prepare a forecast for October usage.

Solution:

Exponential Smoothing ($\alpha = 0.1$)

Given, August forecast $F_{\text{Aug}} = 88\%$

August actual $A_{\text{Aug}} = 89.6\%$

Formula $F_{t+1} = F_t + \alpha (A_t - F_t)$

(A) Forecast for September

$$F_{\text{Sep}} = 88 + 0.1(89.6 - 88) = 88 + 0.16 = 88.16\%$$

(B) Forecast for October (Actual September = 92%)

$$F_{\text{Oct}} = 88.16 + 0.1(92 - 88.16) = 88.16 + 0.384 = 88.54\%$$

Q.2. Compute seasonal relatives for this data using the SA method:

Quarter	Year 1	Year 2	Year 3	Year 4
1	2	3	7	4
2	6	10	18	14
3	2	6	8	8
4	5	9	15	11

Solution: Average for each quarter

$$Q1: (2+3+7+4)/4 = 16/4 = 4$$

$$Q2: (6+10+18+14)/4 = 48/4 = 12$$

$$Q3: (2+6+8+8)/4 = 24/4 = 6$$

$$Q4: (5+9+15+11)/4 = 40/4 = 10$$

$$\text{Grand average} = (4+12+6+10)/4 = 32/4 = 8.$$

Seasonal relatives (simple averages)

$$SR1 = 4/8 = 0.50, SR2 = 12/8 = 1.50, SR3 = 6/8 = 0.75, SR4 = 10/8 = 1.25$$

So the seasonal relatives are:

Quarter 1: 0.50, Quarter 2: 1.50, Quarter 3: 0.75, Quarter 4: 1.25

Q.3. Monthly sales of a product in March, April, and May are 120, 135, and 150. By using linear regression, calculate the demand for June.

Solution:

Let: March $t = 1$, sales $y = 120$

April $t = 2$, sales $y = 135$

May $t = 3$, sales $y = 150$

$$y = a + bt$$

Compute means:

$$\bar{t} = (1 + 2 + 3)/3 = 2$$

$$\bar{y} = (120 + 135 + 150)/3 = 405/3 = 135.$$

2. Find slope b

$$b = \frac{\sum(t - \bar{t})(y - \bar{y})}{\sum(t - \bar{t})^2}$$

Centered data:

- $t = 1: (t - \bar{t}) = -1, (y - \bar{y}) = 120 - 135 = -15, \text{ product} = 15$
- $t = 2: (0, 0, 0)$
- $t = 3: (1, 15, 15)$

$$\sum(t - \bar{t})(y - \bar{y}) = 15 + 0 + 15 = 30.$$

$$\sum(t - \bar{t})^2 = (-1)^2 + 0^2 + 1^2 = 2.$$

$$\text{So } b = 30/2 = 15.$$

Now intercept a

$$a = \bar{y} - b\bar{t} = 135 - 15(2) = 135 - 30 = 105$$

So the regression line is:

$$\hat{y} = 105 + 15t$$

Forecast June (Next Month)

June corresponds to $t = 4$.

$$\hat{y}_{June} = 105 + 15(4) = 105 + 60 = 165$$

Estimated demand for June = 165 units.

Q.4. Use trend-adjusted smoothing with $\alpha = .3$ and $\beta = .2$ to smooth the new account given data. What is the forecast for period 6?

Period	New account
1	200
2	214
3	211
4	228
5	235

Solution:

(4) Use trend-adjusted exponential smoothing with $\alpha = 0.3$, $\beta = 0.2$

Let, L_t = level, T_t = trend, $F_{t+1} = L_t + T_t$.

Take initial values at Period 1.

$$L_1 = A_1 = 200, T_1 = 0$$

After

Period 2,

$$t = 2, A_2 = 214$$

$$L_2 = \alpha A_2 + (1-\alpha)(L_1 + T_1)$$

$$= (0.3 \times 214) + 0.7(200 + 0)$$

$$= 204.2$$

$$T_2 = \beta(L_2 - L_1) + (1-\beta)T_1$$

$$= 0.2(204.2 - 200) + (0.8 \times 0)$$

$$= 0.84$$

$$F_{2+1} = F_3 = 204.2 + 0.84 = 205.04$$

After

Period 3, $t = 3, A_3 = 211$

$$L_3 = 0.3(211) + 0.7(204.2 + 0.84) = 206.828$$

$$T_3 = 0.2(206.828 - 204.2) + 0.8(0.84) = 1.0156$$

$$F_4 = 206.828 + 1.0156 = 207.844$$

After Period 4, $t = 4, A_4 = 228$

$$L_4 = 0.3(228) + 0.7(206.828 + 1.0156)$$

$$= 213.40$$

$$T_4 = 0.2(213.40 - 206.828) + 0.8(1.0156) \\ = 2.4645$$

$$F_5 = 213.40 + 2.4645 = 215.8645$$

After ~~6th~~ → Period 5, $t = 5$, $A_5 = 235$

$$L_5 = 0.3(235) + 0.7(213.40 + 2.4645) \\ = 220.4061$$

$$T_5 = 0.2(220.4061 - 213.40) + 0.8(2.4645) \\ = 3.4503$$

$$\therefore F_6 = 220.4061 + 3.4503 \\ = 223.8564$$

$\therefore$ Therefore the trend-adjusted forecast for period 6 is, $223.8564 \approx 224$ new account.

Q.5.

Assume a previous forecast, including a trend of 110 units, a previous trend estimate of 10 units, an alpha of .20, and a delta of .30. If actual demand turned out to be 115 rather than the forecast 110, calculate the forecast for the next period.

SOLUTION

The actual A_{t-1} is given as 115. Therefore,

$$F_t = FIT_{t-1} + \alpha(A_{t-1} - FIT_{t-1}) \\ = 110 + .2(115 - 110) = 111.0$$

$$T_t = T_{t-1} + \delta(F_t - FIT_{t-1}) \\ = 10 + .3(111 - 110) = 10.3$$

$$FIT_t = F_t + T_t = 111.0 + 10.3 = 121.3$$

If, instead of 121.3, the actual turned out to be 120, the sequence would be repeated and the forecast for the next period would be

$$F_{t+1} = 121.3 + .2(120 - 121.3) = 121.04$$

$$T_{t+1} = 10.3 + .3(121.04 - 121.3) = 10.22$$

$$FIT_{t+1} = 121.04 + 10.22 = 131.26$$